1. Configuration of DNS Server

What is DNS?

The **Domain Name System (DNS)** translates human-readable domain names (e.g., www.microsoft.com) into IP addresses (e.g., 13.77.161.179).

Role of DNS in Windows Server

- Resolves names for ADDS (Active Directory depends heavily on DNS).
- Supports dynamic updates from DHCP and clients.
- Acts as a central part of network services infrastructure.

DNS Server Installation

Steps to install the DNS Server role:

1.

Open **Server Manager**.

2. Go to **Add Roles and Features**.
3. Choose **DNS Server** under **Server Roles**.
4. Proceed through the wizard and install.
5. After installation, configure zones and records.

Types of DNS Zones

- **Primary Zone**: Contains the read-write copy of zone data.
- **Secondary Zone**: Read-only copy, used for load balancing and redundancy.
- **Stub Zone**: Contains only name server (NS), Start of Authority (SOA), and A records to forward queries.

🌐 Zone Types by Scope

- **Forward Lookup Zone**: Resolves names to IPs.
- **Reverse Lookup Zone**: Resolves IPs to names (PTR records).

📄 DNS Records

- **A (Host)**: Maps name to IPv4 address.
- **AAAA**: Maps name to IPv6 address.
- **PTR**: Used in reverse lookup.
- **NS**: Defines authoritative name servers.

- **SOA**: Contains zone-wide info like serial numbers.
- **CNAME**: Alias of one name to another.

□ Dynamic DNS (DDNS)

- Automatically updates DNS records when a client's IP changes.
- Supported with DHCP integration.

□ 2. Install and Configure DHCP

□ What is DHCP?

Dynamic Host Configuration Protocol (DHCP) dynamically assigns IP addresses and other network configuration settings to clients.

󰞵 Benefits of DHCP

- Simplifies IP address management.
- Reduces configuration errors.
- Ensures efficient IP address usage.

󰞵 DHCP Installation Steps

1. Open **Server Manager**.
2. Go to **Add Roles and Features**.
3. Select **DHCP Server** under **Server Roles**.

4. Complete the wizard.
5. After install, use **DHCP Management Console** to configure scopes and options.

🔹 Post-Installation

- **Authorize** DHCP in AD DS.
- Create DHCP **scopes**.
- Configure **scope options** (e.g., default gateway, DNS server).

🔹 3. Manage and Maintain DHCP

🔹 DHCP Scopes

A **scope** is a range of IP addresses that the server can lease.

- Define subnet mask, lease duration, gateway, and DNS.

Reservations

- Assigns a fixed IP to a specific MAC address.
- Useful for printers, servers, or VoIP phones.

Exclusions

- Exclude IPs within a scope range to prevent them from being leased.

Lease Management

- **Lease duration**: Defines how long an IP is assigned.
- Admins can view, delete, and renew leases via the console or PowerShell.

DHCP Policies

- Apply different settings based on device type or MAC.
- Useful for assigning different options to different clients.

Monitoring

- **Event Viewer** for DHCP logs.
- **DHCP Audit Logging** for lease activities.

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4. Implement and Maintain IP Address Management (IPAM)

What is IPAM?

IP Address Management (IPAM) provides a central console to manage IP address spaces, DNS, and DHCP servers.

IPAM Features

- Discover and monitor DNS and DHCP servers.
- Track IP address usage.
- Audit configuration changes.
- Automatically detect IP address conflicts.

IPAM Deployment Methods

- **Manual**: Admin manually adds servers.
- **GPO-based**: Automates provisioning via Group Policy.

IPAM Tasks

- Manage IP address blocks and ranges.
- Detect and correct misconfigurations.
- Delegate IP address space administration.
- Track historical lease and IP assignments.

🔧 Common IPAM Tools

- IP Address Space Management (IPAM console).
- PowerShell modules (e.g., Get-IpamAddress, Add-IpamBlock).
- Integration with DNS and DHCP roles.